

How many pieces does a rolling-shutter needle break into?

Companion to *Falling Forward Through Time* (art_da3u)

Setting. A thin radial needle of length $R = 1$ spins about its hub; a rolling shutter reads row y (measured from the hub row) at a time proportional to y , so row y sees the needle at angle $\theta(y) = M + ey$, where $e = 2\pi kR/H$ (k turns per frame, frame height H ; e is Kepler's eccentricity, not Euler's number, which never appears). Row y meets the needle iff $r = y/\sin\theta(y) \in [0, 1]$, so the visible rows are

$$S = \{y \in [-1, 1] : 0 \leq y/\sin(M + ey) \leq 1\} \cup \{0\}.$$

On S the image point $x = y \cot \theta(y)$ is continuous, and distinct components of S occupy separated bands of rows; hence *pieces* = number of connected components of S .

Theorem. *Let M be uniform mod 2π and $e \geq \pi$. Then*

$$\mathbb{E}[\text{pieces}] = \frac{e}{\pi} + \frac{1}{2} + B(e), \quad \frac{1}{2\pi e} \leq B(e) \leq \frac{\pi}{8e}.$$

Proof. **1. Split at the hub.** For $y > 0$ the condition is $\sin(M + ey) \geq y$; for $y = -s < 0$ it is $\sin(es - M) \geq s$. With $S_+(M) = \{y \in (0, 1] : \sin(M + ey) \geq y\}$ we have $S = S_+(-M)^{\text{refl}} \cup \{0\} \cup S_+(M)$. Almost surely $\sin M \neq 0$, and exactly one side reaches $y = 0$ (the $+$ side iff $\sin M > 0$). Hence $\text{pieces} = 1 + A_+ + A_-$, where $A_{\pm}$ counts the components of $S_{\pm}$ not reaching 0.

2. One hump, at most one piece. Write $\theta = M + ey$. Points of S_+ need $\sin \theta \geq y > 0$, so they lie in positive humps $J_m = [2\pi m, 2\pi m + \pi]$, and different humps are separated by rows with $\sin \theta \leq 0 < y$. On a hump $h(y) = \sin(M + ey) - y$ is concave, so $\{h \geq 0\} \cap J_m$ is an interval: each hump gives at most one component.

3. Exact count. A hump whose peak $\pi/2 + 2\pi m$ lies in $(M, M + e]$ gives exactly one component (at the peak $\sin \theta = 1 \geq y$), except the hump containing $\theta = M$ with $\sin M > 0$, which is glued to the hub; its peak lies in $(M, M + e]$ iff $M \bmod 2\pi \in [0, \pi/2]$ (using $e \geq \pi/2$). The only other hump that can contribute is the partial hump at the end $y = 1$, whose peak lies at $M + e + \delta$, $\delta \in (0, \pi/2)$; with $t = \text{peak} - \theta$ and $1 - y = (t - \delta)/e$ it contributes iff

$$\exists t \in [\delta, \pi/2] : 1 - \cos t \leq (t - \delta)/e \quad (\text{the end bonus}).$$

With $P_{\pm}$ the number of peaks in the window of length e on each side, and since $[M \bmod 2\pi \in [0, \frac{\pi}{2}]] + [-M \bmod 2\pi \in [0, \frac{\pi}{2}]] = [\cos M > 0]$, for every M :

$$\text{pieces} = 1 + P_+ + P_- - [\cos M > 0] + \text{bonus}_+ + \text{bonus}_-.$$

4. Average. A lattice $\pi/2 + 2\pi\mathbb{Z}$ in a window of length e with uniform offset has mean count $e/2\pi$, so $\mathbb{E}P_{\pm} = e/2\pi$; and $\Pr(\cos M > 0) = \frac{1}{2}$. Thus $\mathbb{E}[\text{pieces}] = e/\pi + \frac{1}{2} + B(e)$ with $B(e) = \Pr(\text{bonus}_+) + \Pr(\text{bonus}_-) \geq 0$, where δ is uniform on an interval of length 2π .

5. Bounds on the bonus. *Upper:* for $|t| \leq \pi/2$, $1 - \cos t = 2\sin^2(t/2) \geq 2t^2/\pi^2$, so a bonus requires $\frac{2}{\pi^2}t^2 - \frac{t}{e} + \frac{\delta}{e} \leq 0$ for some t , i.e. discriminant $\frac{1}{e^2} - \frac{8\delta}{\pi^2 e} \geq 0$, i.e. $\delta \leq \pi^2/(8e)$. So $\Pr(\text{bonus}_{\pm}) \leq \pi/(16e)$. *Lower:* $1 - \cos t \leq t^2/2$; if $\delta \leq 1/(2e)$ then $t = 1/e \in [\delta, \pi/2]$ satisfies $t^2/2 \leq (t - \delta)/e$. So $\Pr(\text{bonus}_{\pm}) \geq 1/(4\pi e)$. Summing both sides gives $\frac{1}{2\pi e} \leq B(e) \leq \frac{\pi}{8e}$. $\square$

Remark. Since $1 - \cos t = t^2/2 + O(t^4)$, the lower bound is asymptotically tight: $B(e) = \frac{1}{2\pi e} + O(e^{-2})$ (proof sketched, not written in full).

Numerical check (200 000 random phases per e ; the per-hump count of step 3 agrees with direct pixel counting):

e	$\mathbb{E}[\text{pieces}]$	$e/\pi + \frac{1}{2}$	$B(e)$	$eB(e)$
3	1.5079	1.4549	+0.0530	0.159
7	2.7501	2.7282	+0.0220	0.154
20	6.8754	6.8662	+0.0092	0.185
40	13.2369	13.2324	+0.0045	0.180
80	25.9675	25.9648	+0.0027	0.217

Standard errors ≈ 0.001 in the mean (± 0.08 in eB at $e = 80$); values are consistent with $eB(e) \rightarrow 1/(2\pi) \approx 0.159$. The theorem's bounds assume $e \geq \pi$ (the $e = 3$ row is outside them).

In words. Each half-turn of the needle per frame height adds one piece above the hub and one below; the hub keeps one extra piece on the side the needle starts on (the $+1 - \frac{1}{2}$); and rarely a sliver catches the top or bottom row just before its peak (the $1/(2\pi e)$).